

NITA DCA Part 1

Module 1: Introduction to Computer (10 Hours)

Course: CCC 90-Hour DCA | **Module duration:** 10 hours

Learning Objectives

After this module you will be able to:

- Explain what a computer is, its applications, and types
- Describe the evolution of computers through five generations
- Identify hardware and software components (CPU, memory, I/O, storage)
- Differentiate data, information, and files
- Understand basic number systems (binary, decimal)
- Recognize modern IT gadgets (smartphone, tablet, IoT) and mobile apps
- Follow safe startup and shutdown procedures

1. Introduction to Computers

Computer = electronic device that accepts **input**, processes it using **programs**, stores **data**, and gives **output**.

Applications (व्यवहारिक उपयोग)

- Education, banking, hospitals, offices, e-governance
- Internet, email, online forms, DigiLocker, UPI payments
- CCC and government job exams test basic computer awareness

Types of Computers

Type	Size	Use
Microcomputer (PC, laptop)	Small	Home, office, institute
Minicomputer	Medium	Departments, servers
Mainframe	Large	Banks, railways
Supercomputer	Very large	Weather, research, AI

2. Evolution of Computers — Five Generations

Generation	Period	Technology	Key features	Examples
First (1G)	1940s–1950s	Vacuum tubes	Very large, slow, high power	ENIAC, UNIVAC
Second (2G)	1950s–1960s	Transistors	Smaller, faster, more reliable	IBM 1401

Third (3G)	1960s–1970s	Integrated circuits (ICs)	Multiprogramming, minicomputers	IBM System/360
Fourth (4G)	1970s–present	Microprocessors (VLSI)	Personal computers, GUI, networks	Intel PCs, laptops
Fifth (5G)	Present & future	AI, parallel processing, ULSI	Smart devices, cloud, IoT, voice assistants	Smartphones, supercomputers

Exam tip: Each generation is defined by its **hardware technology**, not by internet speed or mobile networks (5G mobile ≠ fifth computer generation).

3. IT Gadgets and Mobile Computing

Smartphones

- Pocket computer with **touchscreen, camera, GPS, mobile OS** (Android, iOS)
- **Mobile apps** = application software installed from Play Store / App Store
- Uses: calls, messaging, banking (UPI), learning apps, government services (mAadhaar, UMANG)

Tablets

- Larger touchscreen than phone; lighter than laptop
- Good for reading, video classes, note-taking with stylus
- Examples: iPad, Samsung Galaxy Tab, budget Android tablets

IoT (Internet of Things) Devices

- Everyday objects connected to the internet and controlled remotely
- Examples: smart bulbs, CCTV cameras, fitness bands, smart TVs, home assistants
- Need Wi-Fi and often a **mobile app** for setup and control

Other Common Gadgets

Gadget	Function
Laptop	Portable PC for study and office work
Pen drive	Portable USB storage
External HDD/SSD	Backup and large file storage
Webcam / Headset	Online classes and video calls
Printer / Scanner	Hard copy output and digitizing documents

4. Components of a Computer System



Storage

Hardware (tangibles — छू सकते हैं)

Component	Function	Examples
CPU	Brain — executes instructions	Intel Core i3/i5, AMD Ryzen
RAM	Temporary memory while working	4 GB, 8 GB, 16 GB
ROM	Permanent boot instructions	BIOS chip
Hard Disk / SSD	Permanent storage	256 GB, 512 GB, 1 TB
Motherboard	Connects all parts	—
Input devices	Send data to computer	Keyboard, mouse, scanner, microphone
Output devices	Show/print results	Monitor, printer, speaker
VDU	Visual Display Unit = Monitor	LCD, LED

Software (intangible — programs)

- **System software:** Operating System (Windows, Linux, Android)
- **Application software:** Word processor, spreadsheet, browser, mobile apps
- **Utility software:** Antivirus, disk cleanup

Firmware: Software stored in hardware (e.g. BIOS).

5. Central Processing Unit (CPU)

CPU has three parts:

1. **ALU** (Arithmetic Logic Unit) — math and comparisons
2. **CU** (Control Unit) — directs operations
3. **Registers** — tiny fast storage inside CPU

Speed measured in **GHz** (gigahertz). More GHz + more cores = faster processing.

6. Memory

Memory	Full Form	Volatile?	Use
RAM	Random Access Memory	Yes (clears on shutdown)	Running programs
ROM	Read Only Memory	No	Boot-up
Cache	—	Yes	Very fast buffer for CPU
Virtual memory	—	Uses disk as extra RAM	When RAM is full

1 Byte = 8 bits
1 KB = 1024 bytes | 1 MB = 1024 KB | 1 GB = 1024 MB | 1 TB = 1024 GB

7. Input and Output Devices

Input

- **Keyboard:** QWERTY layout; function keys F1–F12; shortcuts Ctrl, Alt, Shift
- **Mouse:** Left click (select), right click (menu), double-click (open), scroll wheel
- **Scanner:** Converts paper → digital image
- **Webcam / Microphone:** Video calls, recording
- **Touchscreen:** Input on phones and tablets

Output

- **Monitor:** Resolution e.g. 1920×1080 (Full HD)
- **Printer:** Inkjet (colour photos), Laser (fast text), Dot matrix (bills)
- **Speaker / Headphones:** Audio output

8. Storage Devices

Device	Type	Speed	Portability
HDD	Magnetic disk	Slower	Internal/external
SSD	Flash chips	Fast	Internal/external
Pen drive (USB)	Flash	Medium	Highly portable
CD/DVD	Optical	Slow	Portable
Memory card	Flash	Medium	Camera, phone

9. Data and Information

- **Data:** Raw facts (numbers, text, images) — e.g. 85, 90, 78
- **Information:** Processed meaningful data — e.g. "Average marks = 84.3"

File: Named collection of data stored on disk.
Folder (Directory): Container for files and subfolders.

Common extensions: .txt , .docx , .odt , .xlsx , .ods , .pptx , .odp , .pdf , .jpg , .mp4 , .apk

10. Number Systems (Basic)

System	Base	Digits	Use
Decimal	10	0–9	Daily life

Binary	2	0, 1	Computer internal
Hexadecimal	16	0–9, A–F	Memory addresses

Binary example: 1010 = 1×8 + 0×4 + 1×2 + 0×1 = **10** (decimal)

11. Connecting a Computer (Lab Theory)

Steps to set up a desktop PC

1. Place CPU, monitor, keyboard, mouse on stable surface
2. Connect monitor cable (HDMI/VGA) to CPU
3. Connect keyboard and mouse to USB ports
4. Connect power cables; switch on UPS first (if used), then CPU and monitor
5. Wait for OS logo to appear

Safe shutdown

1. Save all work
 2. Start → Power → Shut down (never unplug while running)
 3. Switch off UPS after shutdown completes
-

12. Important Terms (Exam Vocabulary)

Term	Meaning
Hardware	Physical parts of computer
Software	Programs and instructions
Booting	Starting the computer
Cold boot	Start from off state
Warm boot	Restart without full power off
GUI	Graphical User Interface (icons, mouse)
CLI	Command Line Interface (text commands)
Multitasking	Running multiple programs at once
Mobile app	Application software on smartphone/tablet
IoT	Devices connected to internet for remote control
Virus	Harmful program that spreads
Backup	Copy of data for safety

Lab Tasks (Hands-on)

1. **Identify parts:** Label CPU, monitor, keyboard, mouse, USB ports on your lab PC.
 2. **Storage check:** Open File Manager / This PC → note C: drive total size and free space.
 3. **Device list:** List 3 input and 3 output devices you use daily (include phone/tablet).
 4. **File types:** Create or locate 5 files with extensions `.txt` , `.docx` , `.xlsx` , `.pptx` , `.jpg` — observe icons.
 5. **Binary convert:** Convert decimal 13, 25, 31 to binary (show working).
 6. **Generations chart:** Draw a table of 5 computer generations with technology used in each.
 7. **Mobile apps:** On a smartphone, list 5 apps — classify each as system, utility, or application software.
 8. **IoT survey:** Name 3 IoT or smart gadgets at home or college and what they connect to.
-

Chapter Summary Checklist

- ☐ I can name 5 hardware and 5 software examples
 - ☐ I know all 5 generations and their key technology
 - ☐ I know RAM vs ROM vs hard disk
 - ☐ I can explain input vs output devices
 - ☐ I know KB, MB, GB conversions
 - ☐ I can describe smartphone, tablet, and IoT basics
 - ☐ I can connect and safely shut down a PC
-

Quick MCQs (10 Questions)

1. Which generation of computers used vacuum tubes?

- a) Second
- b) Third
- c) First
- d) Fourth

2. The brain of the computer is:

- a) RAM
- b) CPU
- c) Hard disk
- d) Monitor

3. Which memory is lost when power is switched off?

- a) ROM
- b) RAM
- c) BIOS
- d) Firmware

4. A smartphone app downloaded from Play Store is an example of:

- a) Hardware
- b) System software
- c) Application software
- d) Firmware

5. IoT devices are best described as:

- a) Computers without keyboards
- b) Objects connected to the internet for monitoring/control
- c) Only supercomputers
- d) Optical storage devices

6. Binary number **1101** in decimal is:

- a) 11
- b) 13
- c) 15
- d) 9

7. Fourth generation computers are based on:

- a) Vacuum tubes
- b) Transistors
- c) Microprocessors
- d) Mechanical gears

8. Which is an output device?

- a) Scanner
- b) Keyboard
- c) Microphone
- d) Printer

9. 1 GB equals approximately:

- a) 1000 bytes
- b) 1024 MB
- c) 1024 KB
- d) 8 bits

10. Processed meaningful data is called:

- a) Input
- b) Output
- c) Information
- d) Hardware

MCQ Answers

1-c | 2-b | 3-b | 4-c | 5-b | 6-b | 7-c | 8-d | 9-b | 10-c

Module 2: Operating System (10 Hours)

Course: CCC 90-Hour DCA | **Module duration:** 10 hours

Learning Objectives

After this module you will be able to:

- Describe functions of an operating system
 - Navigate Windows desktop, Start menu, taskbar, and Settings
 - Create, rename, copy, move, and delete files and folders
 - Use File Explorer effectively
 - Install and remove (uninstall) programs safely
 - Understand Linux/Ubuntu basics and open-source software
 - Use built-in utilities (Notepad, Calculator, Paint, Snipping Tool)
 - Apply basic security habits (lock screen, user accounts, updates)
-

1. Introduction to Operating System

OS = software that manages hardware and provides interface for users and applications.

Functions of OS

1. **Process management** — runs and schedules programs
2. **Memory management** — allocates RAM to applications
3. **File management** — folders, permissions, storage
4. **Device management** — drivers for printer, keyboard, etc.
5. **User interface** — GUI (icons, mouse) or CLI (text commands)

Popular Operating Systems

OS	Type	Use
Windows 10/11	GUI	Most offices, institutes, CCC labs
Linux (Ubuntu)	GUI + CLI	Servers, developers, free PCs
macOS	GUI	Apple computers
Android / iOS	Mobile GUI	Smartphones and tablets

This module focuses on **Windows GUI**, with **Linux/Ubuntu** and **open source** introduced for CCC syllabus coverage.

2. Open Source Software and Linux/Ubuntu

Open Source

- **Source code** is publicly available — anyone can view, modify, and share (under a license)
- Usually **free to download** and use; community or foundation maintains it
- Examples: **Linux, LibreOffice, Firefox, VLC Media Player**

Closed Source (Proprietary)

- Source code is private; only the vendor can modify
- Examples: Microsoft Windows, MS Office (commercial license)

Linux — Introduction

- **Linux** = open-source OS kernel; many **distributions (distros)** package it with apps
- **Ubuntu** = popular beginner-friendly Linux distro with GUI (similar tasks to Windows)

- Used on servers worldwide; also runs on old PCs via dual-boot or live USB

Windows vs Ubuntu (quick comparison)

Task	Windows	Ubuntu (Linux)
File manager	File Explorer	Files (Nautilus)
App store	Microsoft Store	Ubuntu Software / Snap Store
Settings	Settings (Win+I)	Settings app
Terminal	Command Prompt / PowerShell	Terminal (bash)
Install apps	.exe / Microsoft Store	.deb / Snap / apt
Office suite	MS Office	LibreOffice (pre-installed)

Trying Ubuntu (lab option)

1. Download Ubuntu ISO from ubuntu.com
 2. Create bootable USB (Rufus) **or** use virtual machine (VirtualBox)
 3. Explore desktop, Files app, Firefox, LibreOffice, Terminal
-

3. Desktop and Taskbar (Windows)

Desktop elements

- **Icons:** Shortcuts to programs/files
- **Taskbar:** Bottom bar — Start, pinned apps, system tray, clock
- **System tray:** Network, volume, battery, date/time
- **Wallpaper:** Background image (right-click desktop → Personalize)

Start Menu

- Search box — type app name
- Pinned apps — Word, Excel, Settings
- Power button — Shut down, Restart, Sleep

Keyboard shortcuts (must know)

Shortcut	Action
Win	Open Start menu
Win + E	File Explorer
Win + D	Show desktop
Win + L	Lock computer
Win + I	Settings
Alt + Tab	Switch between open windows

Alt + F4	Close active window
Ctrl + C / V / X	Copy / Paste / Cut
Ctrl + Z	Undo
Ctrl + S	Save
F2	Rename selected item
Delete	Send to Recycle Bin
Shift + Delete	Permanent delete

4. File Explorer

Open: Win + E or taskbar folder icon

Parts of File Explorer

- **Navigation pane** (left): Quick access, This PC, drives
- **Address bar:** Current path e.g. `C:\Users\Student\Documents`
- **Ribbon:** Home, Share, View tabs
- **File list:** Name, Date modified, Type, Size

Views

- **Icons / List / Details / Tiles** — View tab
- **Sort** by name, date, size, type
- **Search box** — find files by name inside folder

Drives

- **C:** — Usually Windows and programs
 - **D:** or others — Data, DVD, USB when connected
-

5. File and Folder Operations

Create folder

1. Open desired location in File Explorer
2. Right-click → New → Folder
3. Type name → Enter

Copy / Move

- **Copy:** Ctrl+C → go to destination → Ctrl+V (original stays)
- **Move (Cut):** Ctrl+X → destination → Ctrl+V
- **Drag-drop:** Drag with left mouse; hold **Shift** to move, **Ctrl** to copy

Rename

- Select item → F2 → type new name → Enter

- Do not change file extension (.docx) unless you know the type

Delete and Restore

- Delete → Recycle Bin (can restore)
- Empty Recycle Bin → permanent delete
- Shift + Delete → skip Recycle Bin

File naming rules (Windows)

- Invalid characters: \ / : * ? " < > |
 - Extensions identify type: report.docx , data.xlsx
 - Case usually not sensitive: File.txt = file.txt
-

6. Installing and Removing Programs

Installing software (Windows)

Method 1 — Installer (.exe / .msi)

1. Download from official website only (avoid unknown links)
2. Double-click installer → follow wizard (Next → I Agree → Install)
3. Choose install location if asked (default is usually fine)
4. Finish → launch from Start menu

Method 2 — Microsoft Store

1. Open Microsoft Store from Start
2. Search app → Get / Install
3. Updates often automatic

Method 3 — Open-source on Windows

- Download LibreOffice, Firefox, VLC from official sites — same wizard install

Ubuntu equivalent: Ubuntu Software → search → Install, or Terminal: `sudo apt install libreoffice`

Removing / Uninstalling software (Windows)

Settings method (recommended)

1. Win + I → **Apps** → **Installed apps** (or Apps & features)
2. Search program name → ⋮ → **Uninstall**
3. Follow prompts; restart if asked

Control Panel method (classic)

1. Control Panel → Programs → Uninstall a program
2. Select program → Uninstall

Ubuntu equivalent: Ubuntu Software → Installed → Remove, or `sudo apt remove packagename`

Safe installation habits

- Install only needed software; avoid pirated/cracked copies
 - Uncheck bundled extra toolbars or unwanted apps during setup
 - Keep Windows Update enabled for security patches
 - Remove unused programs to free disk space
-

7. Control Panel and Settings

Settings (Win + I): Modern — System, Devices, Network, Personalization, Accounts, Update

Control Panel: Classic — Programs, Devices and Printers, User Accounts

Common tasks

Task	Path
Change display	Settings → System → Display
Add printer	Settings → Bluetooth & devices → Printers
Uninstall program	Settings → Apps → Installed apps
Create user	Settings → Accounts → Family & other users
Date/time	Taskbar clock → Adjust date/time
Windows Update	Settings → Windows Update

8. Built-in Utilities

Program	Use	Open
Notepad	Plain text	Start → Notepad
WordPad	Rich text	Start → WordPad
Calculator	Standard / Scientific	Start → Calculator
Paint	Simple drawing	Start → Paint
Snipping Tool	Screenshot	Start → Snipping Tool
Command Prompt	Text commands	Start → cmd
Task Manager	Running programs, performance	Ctrl+Shift+Esc

Task Manager uses

- End frozen program (End task)
 - Check CPU/RAM usage (Performance tab)
 - Startup programs (Startup tab)
-

9. User Accounts and Security

- **Administrator:** Full control — install software, change settings
- **Standard user:** Limited — safer for daily use
- **Password / PIN:** Lock screen protection
- **Windows Update:** Keep OS patched — Settings → Windows Update

Good habits

- Lock PC when away (Win + L)
 - Do not share password
 - Regular backup to USB or cloud
 - Use antivirus (Windows Defender built-in)
-

10. Compression and Shortcuts

Zip (compress)

- Right-click folder → Send to → Compressed (zipped) folder
- Reduces size for email/upload

Desktop shortcut

- Right-click program → Send to → Desktop (create shortcut)
 - Or right-click desktop → New → Shortcut → browse to .exe
-

Lab Tasks

1. Create folder structure: `Documents\CCC\Unit2\Notes` and `Documents\CCC\Unit2\Labs`
 2. Create `my-info.txt` in Notes with your name, course, date
 3. Copy to Labs; rename copy to `my-info-backup.txt`
 4. Practice shortcuts: Win+E, Win+D, Win+L, Alt+Tab, F2, Ctrl+C/V/X
 5. Take screenshot with Snipping Tool; save as PNG in Labs
 6. Open Task Manager — note CPU and memory usage %
 7. Create zip of Labs folder; compare size with original folder
 8. **Install:** Install LibreOffice or Firefox from official site (or use Store); note install steps
 9. **Uninstall:** Remove a practice app via Settings → Apps → Uninstall (teacher-approved app only)
 10. **Linux demo:** Boot Ubuntu live USB or VM — open Files, Firefox, and LibreOffice Writer; compare with Windows
-

Summary Checklist

- ☐ I can explain five functions of an OS
- ☐ I know open source vs closed source with examples
- ☐ I can name basic Ubuntu apps equivalent to Windows tools
- ☐ I can navigate File Explorer and manage files/folders
- ☐ I can install and uninstall a program safely
- ☐ I know 10+ keyboard shortcuts

- ☐ I can use Notepad, Calculator, Snipping Tool, Task Manager
-

Quick MCQs (10 Questions)

1. Which is an open-source operating system?

- a) Windows 11
- b) macOS
- c) Ubuntu Linux
- d) iOS

2. Primary function of an operating system is:

- a) Only playing games
- b) Managing hardware and providing user interface
- c) Replacing the CPU
- d) Storing data permanently without software

3. Shortcut to open File Explorer in Windows:

- a) Win + E
- b) Win + P
- c) Ctrl + E
- d) Alt + E

4. To move a file (not copy) using keyboard:

- a) Ctrl+C then Ctrl+V
- b) Ctrl+X then Ctrl+V
- c) Shift+Delete
- d) F2

5. Recommended way to uninstall a program in Windows 10/11:

- a) Delete folder from Program Files only
- b) Settings → Apps → Installed apps → Uninstall
- c) Rename the .exe file
- d) Empty Recycle Bin

6. LibreOffice is an example of:

- a) Hardware
- b) Open-source application suite
- c) Virus
- d) CPU type

7. Shift + Delete does what?

- a) Sends file to Recycle Bin
- b) Permanently deletes without Recycle Bin
- c) Renames file
- d) Copies file

8. Win + L is used to:

- a) Open LibreOffice
- b) Lock the computer
- c) Log off permanently
- d) Open Linux terminal

9. Ubuntu's graphical file manager is commonly called:

- a) File Explorer
- b) Finder
- c) Files (Nautilus)
- d) Notepad

10. Task Manager is opened by:

- a) Ctrl+Alt+Del only
- b) Ctrl+Shift+Esc
- c) Win+T
- d) F12

MCQ Answers

1-c | 2-b | 3-a | 4-b | 5-b | 6-b | 7-b | 8-b | 9-c | 10-b

Module 3: Word Processing (12 Hours)

Course: CCC 90-Hour DCA | **Module duration:** 12 hours

Learning Objectives

After this module you will be able to:

- Create, save, open, and print documents in MS Word and LibreOffice Writer
 - Apply character and paragraph formatting
 - Use bullets, numbering, borders, tables, and images
 - Insert headers, footers, and page numbers
 - Use find/replace, spell check, and styles
 - Prepare professional **letters** and **resumes**
 - Perform basic mail merge
-

1. Introduction to Word Processing

A **word processor** creates and edits text documents with formatting, unlike plain Notepad.

Software	Developer	Default extension	Open with
MS Word	Microsoft	.docx	Word, Writer (partial)
LibreOffice Writer	The Document Foundation (open source)	.odt	Writer, Word (partial)

CCC note: Skills transfer between Word and Writer — menus differ slightly but concepts are the same.

Open a blank document

- **Word:** Start → Microsoft Word → Blank document
- **Writer:** Start → LibreOffice Writer (or search Writer)

Screen layout (both apps)

Area	Purpose
Title bar	Document name, window controls
Ribbon / Menus	Home, Insert, Layout/Page, Review, View
Ruler	Margins, indents
Document area	Typing space
Status bar	Page count, word count, zoom

Save

- **Ctrl + S** — first time: choose location and filename
 - **Save As** — new name, `.pdf` , or `.odt` / `.docx` format
 - **Word:** File → Save As → PDF
 - **Writer:** File → Export as PDF
-

2. MS Word vs LibreOffice Writer — Feature Parity

Task	MS Word	LibreOffice Writer
Bold / Italic	Ctrl+B / Ctrl+I	Same
Alignment	Home → alignment	Home (toolbar) → alignment
Line spacing	Home → Line spacing	Format → Paragraph → Line spacing
Table	Insert → Table	Insert → Table
Picture	Insert → Pictures	Insert → Image
Header/Footer	Insert → Header & Footer	Insert → Header/Footer
Page break	Ctrl+Enter	Ctrl+Enter
Find/Replace	Ctrl+F / Ctrl+H	Ctrl+F / Ctrl+H
Spell check	Review → Spelling (F7)	Tools → Spelling (F7)
Styles	Home → Styles	Styles sidebar (F11)
Mail merge	Mailings tab	Tools → Mail Merge Wizard

3. Basic Text Entry and Editing

- **Enter** — new paragraph | **Shift + Enter** — line break in same paragraph
- **Backspace** — delete left | **Delete** — delete right
- **Select:** drag mouse, Ctrl+A (all), Shift+arrows
- **Undo / Redo:** Ctrl+Z / Ctrl+Y (Writer: Ctrl+Shift+Z for redo)

4. Character Formatting

Tool	Shortcut	Use
Bold	Ctrl+B	Headings, emphasis
Italic	Ctrl+I	Titles, foreign words
Underline	Ctrl+U	Underline
Font size	—	12 pt body; 14–16 headings
Font color / Highlight	—	Emphasis
Clear formatting	—	Reset to default

Fonts: Arial, Times New Roman, Calibri (Word); Liberation Serif/Sans (Writer)

5. Paragraph Formatting

Setting	Notes
Alignment	Left, Center, Right, Justify (formal letters, essays)
Line spacing	1.0, 1.5, 2.0 — resumes often use 1.0–1.15
Before/After spacing	Space between paragraphs
Indent	First-line indent for essays; hanging indent for references
Bullets / Numbering	Lists and outlines

6. Page Layout

- **Margins:** Layout (Word) / Format → Page (Writer) — Normal, Narrow, Custom
- **Orientation:** Portrait (letters, resumes) / Landscape (tables, certificates)
- **Size:** A4 (standard India)
- **Columns:** Layout → Columns — newsletters
- **Page Break:** Ctrl+Enter

7. Insert — Tables, Pictures, Headers

Table

Insert → Table → choose rows×columns

Tab = next cell | Table design: borders, shading, header row

Pictures

Insert → Picture/Image → resize; **Wrap Text** (Square, Tight) for text flow

Header, Footer, Page Number

Insert → Header/Footer → type institute name or "Resume"

Insert → Page Number → bottom center (avoid on one-page resume)

8. Find, Replace, Spell Check

- **Find:** Ctrl+F | **Replace:** Ctrl+H
 - **Spelling:** F7 or Review/Tools → Spelling
 - **Word count:** Review → Word Count (Writer: Tools → Word Count)
-

9. Styles and Templates

Styles: Heading 1, Heading 2, Normal — consistent structure for reports

Built-in templates

- **Word:** File → New → search "Resume", "Cover letter", "Formal letter"
 - **Writer:** File → New → Templates → Letters, CV/resume templates
-

10. Practical: Formal Letter

Structure of a formal letter

[Your address – top right or left]

[Date]

[Recipient name & designation]

[Organization address]

Subject: [Bold, one line]

Respected Sir/Madam,

[Body – 2-3 paragraphs, justified]

Yours faithfully,
[Your name]
[Contact: phone, email]

Formatting checklist for letters

- A4, margins 2.5 cm (Normal)
- Font: Times New Roman 12 pt or Arial 11 pt
- Subject line bold; paragraphs justified
- Leave space for signature above your name

Lab reference letter topic

Write to the Principal requesting **study leave** or **fee concession** — one page, proper salutation and subject.

11. Practical: Resume (CV)

Resume sections (one page for freshers)

1. **Name & contact** — phone, email, city (no need for full address in short CV)
2. **Career objective** — 2 lines (optional for CCC level)
3. **Education** — Class, board, year, percentage (table or lines)
4. **Skills** — MS Word, Excel, typing speed, English/Hindi
5. **Certifications** — CCC, NCVRT, etc.
6. **Declaration** — optional one line

Resume formatting tips

- **One page** for students; clean font (Calibri 11, Arial 10)
 - Use **bold** for section headings; avoid colours overload
 - **No photo** unless employer asks (India government forms sometimes require)
 - Export **PDF** before submitting — layout stays fixed
 - Word: File → New → "Simple resume" | Writer: Template → Curriculum vitae
-

12. Mail Merge (Overview)

Purpose: One letter to many people with personalized name/address.

Steps (Word)

1. Mailings → Start Mail Merge → Letters
2. Select Recipients → use Excel list or type new list
3. Insert Merge Fields (Name, Address, City)
4. Preview Results → Finish & Merge → Print or Edit documents

Writer equivalent

Tools → Mail Merge Wizard → follow steps (data source: spreadsheet .ods/.xlsx)

13. Print and Export

- **Print:** Ctrl+P — printer, pages, copies
 - **PDF:** Save As / Export as PDF — submit assignments and resumes
-

Lab Tasks

1. Type a 1-page essay "Uses of Computer" — Heading 1 title, justify, 1.5 spacing
 2. Create **formal letter** to Principal (study leave) — correct layout on A4
 3. Create **one-page resume** using a template; fill your real education and skills; export PDF
 4. Build 4×4 marks table (Name, Roll, Subject, Marks)
 5. Insert image with Square text wrap; add header "NITA Classes" and page numbers
 6. Find & replace one word throughout a document
 7. Repeat letter OR resume in **LibreOffice Writer** — save as `.odt` and export PDF
 8. Optional: Mail merge letter to 5 recipients using small Excel/Calc list
-

Summary Checklist

- ☐ I can work in both Word and Writer for basic tasks
 - ☐ I can format text and paragraphs professionally
 - ☐ I can create a formal letter with subject and salutation
 - ☐ I can create a one-page resume and export PDF
 - ☐ I can insert tables, images, header/footer
 - ☐ I can run spell check and find/replace
 - ☐ I understand basic mail merge steps
-

Quick MCQs (10 Questions)

1. Default LibreOffice Writer file extension is:

- a) .docx
- b) .odt
- c) .xlsx
- d) .pptx

2. Shortcut for Save in Word/Writer:

- a) Ctrl+P
- b) Ctrl+S
- c) Ctrl+N
- d) F7

3. For a formal letter in India, paper size is usually:

- a) Letter
- b) A4
- c) Legal only
- d) A3

4. Mail merge is used to:

- a) Merge two printers
- b) Create personalized bulk letters from one template
- c) Delete duplicate pages
- d) Compress files

5. In Writer, spell check is commonly under:

- a) Tools → Spelling
- b) Insert → Picture
- c) View → Zoom
- d) File → Print

6. Page break shortcut:

- a) Ctrl+Enter
- b) Shift+Enter
- c) Alt+Enter
- d) Ctrl+Break

7. A fresher resume should ideally be:

- a) 5 pages with graphics
- b) One page, clear sections, PDF for submission
- c) Only handwritten
- d) Landscape orientation only

8. Justify alignment means:

- a) Text aligned left only
- b) Both left and right edges straight
- c) Text in centre
- d) Vertical text

9. Which is open-source word processor?

- a) MS Word
- b) LibreOffice Writer
- c) Notepad
- d) Paint

10. Header and footer appear:

- a) Only on first page always
- b) At top and bottom of pages (can vary by section)
- c) Only in PDF
- d) Behind the text always

MCQ Answers

1-b | 2-b | 3-b | 4-b | 5-a | 6-a | 7-b | 8-b | 9-b | 10-b

Module 4: Spreadsheet (12 Hours)

Course: CCC 90-Hour DCA | **Module duration:** 12 hours

Learning Objectives

After this module you will be able to:

- Navigate workbook, sheets, cells, and ranges in Excel and LibreOffice Calc
- Enter and format data (numbers, text, dates, currency)
- Write formulas and use common functions (SUM, AVERAGE, IF, COUNTIF)
- Apply relative and absolute cell references
- Sort, filter, and use conditional formatting
- Create and customize charts
- Set print area, headers, and page setup for worksheets

Focus: Spreadsheet skills in **MS Excel** and **LibreOffice Calc** — not database tools (Access/Base).

1. Introduction to Spreadsheets

A **spreadsheet** organizes data in rows and columns for calculation, analysis, and charts.

Software	Extension	Open source?
MS Excel	.xlsx	No (Microsoft)
LibreOffice Calc	.ods	Yes

Basic terms

- **Workbook** = entire file
- **Worksheet (Sheet)** = tab (Sheet1, Sheet2...)
- **Cell** = intersection of row and column
- **Cell address:** Column letter + row number — **A1, D15**
- **Range:** A1:C10 (rectangle)

Fill handle

Drag small square at cell corner to copy formula or extend series (1,2,3... or Jan, Feb...)

2. MS Excel vs LibreOffice Calc — Parity

Task	MS Excel	LibreOffice Calc
New workbook	Ctrl+N	Ctrl+N
Save	Ctrl+S	Ctrl+S
Format cells	Ctrl+1	Ctrl+1

Sum formula	=SUM(A1:A10)	Same syntax
AutoSum	Home → AutoSum	Sigma Σ on toolbar
Sort/Filter	Data tab	Data menu
Chart	Insert → Chart	Insert → Chart
Sheet tab	Bottom tabs	Bottom tabs
Print preview	Ctrl+P	Ctrl+P
Export PDF	File → Save As PDF	File → Export as PDF

Formula syntax is nearly identical — CCC exams often use Excel notation; it works in Calc.

3. Data Entry and Formatting

Data types

- **Text** — left-aligned (names, roll numbers as text)
- **Numbers** — right-aligned (marks, amounts)
- **Dates** — format dd-mm-yyyy (India)

Home tab / Formatting toolbar

Tool	Use
Number format	General, Number, Currency ₹, Percentage, Date
Decimal places	Increase/decrease
Bold / Borders	Table appearance
Merge & Center	Title across columns
Wrap Text	Multi-line in one cell

Format Cells (Ctrl+1): Number, Alignment, Font, Border, Fill

4. Formulas (always start with =)

Formula	Example	Result
Addition	=A1+B1	Sum of two cells
Subtraction	=A1 - B1	Difference
Multiplication	=A1*B1	Product
Division	=A1/B1	Quotient

Average	=AVERAGE(A1:A10)	Mean
Sum	=SUM(A1:A10)	Total
Count	=COUNT(A1:A10)	Count numbers
Max / Min	=MAX(A1:A10) / =MIN()	Highest / lowest
Percentage	=B1/C1*100	% marks

Cell references

- **Relative:** =B2*C2 — changes when copied down (B3C3, B4C4...)
- **Absolute:** =\$B\$1 — fixed when copying (press **F4** in Excel/Calc)
- **Mixed:** =\$B1 or =B\$1 — lock one part only

Use case: Tax rate in cell B1 (\$B\$1) applied to every row of amounts.

5. Important Functions (Exam Focus)

Function	Syntax	Use
SUM	=SUM(B2:B20)	Total
AVERAGE	=AVERAGE(B2:B20)	Average
COUNT	=COUNT(B2:B20)	Count numeric cells
COUNTA	=COUNTA(A2:A20)	Count non-empty cells
MAX / MIN	=MAX(), =MIN()	Extremes
IF	=IF(C2>=40, "Pass", "Fail")	Condition
COUNTIF	=COUNTIF(B2:B20, ">=40")	Count if condition
SUMIF	=SUMIF(A2:A20, "Delhi", B2:B20)	Sum matching criteria
CONCAT / &	=A2&" "&B2	Join text (Calc: CONCATENATE)
TODAY	=TODAY()	Current date
ROUND	=ROUND(A2, 2)	Round decimals

Nested IF (grades):

=IF(B2>=60, "First", IF(B2>=45, "Second", "Fail"))

6. Sort and Filter

- **Sort:** Data → Sort A to Z / Z to A / Custom sort (multiple columns)
- **Filter:** Data → Filter — dropdown arrows in headers; show only matching rows

- **Remove Duplicates:** Data → Remove Duplicates (Excel); Calc: Data → More filters → Standard filter

Example: Sort students by total marks descending; filter total > 200.

7. Conditional Formatting

Home → Conditional Formatting (Calc: Format → Conditional)

- Highlight Cell Rules — greater than, less than, between
- Color scales — heat map
- Data bars — visual bars in cells
- Icon sets — arrows, traffic lights

Example: Marks < 40 → red fill; marks > 75 → green fill.

8. Charts

Insert → Chart → select data range with labels

Chart type	Best for
Column / Bar	Compare categories (subject-wise marks)
Line	Trends over time (monthly sales)
Pie	Parts of whole (pass/fail proportion)
Area	Volume over time

Steps: Select data including row/column labels → Insert chart → add title, legend, axis labels.

9. Page Setup and Print

- **Print area:** Page Layout → Print Area → Set Print Area
 - **Print titles:** Repeat row 1 (headers) on every printed page
 - **Orientation:** Portrait (tall data) / Landscape (wide tables)
 - **Margins:** Narrow for more columns on one page
 - **Gridlines:** Print gridlines option for clarity
 - **Preview:** Ctrl+P
-

10. Multiple Sheets

- **Add sheet:** + tab or Shift+F11
 - **Rename:** Double-click tab
 - **3D formula:** =SUM(Sheet1:Sheet3!B2) — same cell across sheets
 - **Link:** =Sheet2!A1 — reference another sheet
-

11. Typical CCC Spreadsheet Tasks

Task	Approach
Student marksheet	Formulas: Total, Average, %, Grade (IF)
Fee collection	SUM, date column, receipt number
Attendance %	COUNTA, division, percentage format
Inventory list	Sort by quantity; filter low stock
Simple budget	SUM by category; pie chart of expenses

Lab Tasks

1. Create marks sheet: 10 students, 3 subjects — Total, Average, % (formulas)
2. Add **Pass/Fail** column with `=IF(total>=120,"Pass","Fail")` or per-subject IF
3. Use COUNTIF to count passes (>=40) in each subject
4. Sort by total descending; filter students with total > 200
5. Conditional format: fail marks red, distinction (>75) green
6. Column chart of totals; pie chart of pass/fail count
7. Set print area and repeat header row; print preview
8. Use **absolute reference**: tax rate in \$B\$1 applied to all amount rows
9. Repeat tasks 1–4 in **LibreOffice Calc** — save as `.ods` and export PDF

Summary Checklist

- ☐ I can work in Excel and Calc with same formulas
- ☐ I write SUM, AVERAGE, IF, COUNTIF correctly
- ☐ I use relative and absolute references (F4)
- ☐ I sort, filter, and apply conditional formatting
- ☐ I create column and pie charts
- ☐ I set print titles and print area
- ☐ I know spreadsheet ≠ database (no Access focus in this module)

Quick MCQs (10 Questions)

1. Cell at column D row 10 is:

- a) 10D
- b) D10
- c) D:10
- d) 4J

2. Every formula in Excel/Calc must begin with:

- a) +
- b) =
- c) #
- d) @

3. To keep cell B1 fixed when copying a formula down, use:

- a) B1
- b) \$B\$1
- c) B\$1 only in Excel Mac
- d) #B1

4. Function to count only numeric cells in a range:

- a) COUNTA
- b) COUNT
- c) SUM
- d) LEN

5. LibreOffice Calc default file extension:

- a) .xlsx
- b) .ods
- c) .odt
- d) .mdb

6. `=IF(C2>=40,"Pass","Fail")` returns "Fail" when:

- a) C2 is 50
- b) C2 is 40
- c) C2 is 35
- d) C2 is text "Pass"

7. Best chart to show marks of 5 subjects for one class:

- a) Pie chart of one student
- b) Column/Bar chart
- c) Scatter with no labels
- d) Only line chart for categories

8. Conditional formatting is used to:

- a) Change hardware
- b) Automatically format cells based on rules
- c) Install fonts
- d) Merge sheets

9. Shortcut to open Format Cells dialog:

- a) Ctrl+1
- b) Ctrl+F
- c) Alt+Tab
- d) F5

10. A workbook contains:

- a) Only one cell
- b) One or more worksheets
- c) Only charts

- d) Only macros

MCQ Answers

1-b | 2-b | 3-b | 4-b | 5-b | 6-c | 7-b | 8-b | 9-a | 10-b

Module 5: Presentation (8 Hours)

Course: CCC 90-Hour DCA | **Module duration:** 8 hours

Learning Objectives

After this module you will be able to:

- Create multi-slide presentations in MS PowerPoint and LibreOffice Impress
 - Apply themes, layouts, and consistent design
 - Add and format text, bullets, images, charts, and tables
 - Use slide transitions and object animations appropriately
 - Run slideshow with keyboard navigation and presenter tools
 - Export presentations as PDF, video, or slideshow file
-

1. Introduction to Presentation Software

A **presentation** = sequence of **slides** shown to an audience — classes, seminars, project demos.

Software	Extension	Notes
MS PowerPoint	.pptx	Industry standard in offices
LibreOffice Impress	.odp	Open source; free with Ubuntu/LibreOffice

Open blank presentation

- **PowerPoint:** Start → PowerPoint → Blank Presentation
- **Impress:** Start → LibreOffice Impress

Key terms

- **Slide** = one screen of content
 - **Slide layout** = arrangement of placeholders (title, content, blank)
 - **Theme** = coordinated colours, fonts, backgrounds
 - **Transition** = effect **between** slides
 - **Animation** = effect **on objects within** a slide
-

2. MS PowerPoint vs LibreOffice Impress — Parity

Task	PowerPoint	Impress
------	------------	---------

New slide	Ctrl+M / Home → New Slide	Slide → New Slide
Slide show from start	F5	F5
From current slide	Shift+F5	Shift+F5
Duplicate slide	Right-click → Duplicate	Slide → Duplicate slide
Insert picture	Insert → Pictures	Insert → Image
Transitions	Transitions tab	Slide Transition sidebar
Animations	Animations tab	Custom Animation
Export PDF	File → Save As PDF	File → Export as PDF
Video export	File → Export → Video	File → Export (varies by version)
Direct slideshow file	.ppsx	Configure in slide show settings

3. Views

View	PowerPoint	Use
Normal	Default	Edit slides
Slide Sorter	View → Slide Sorter	Reorder many slides
Reading View	View → Reading View	Preview full screen
Slide Show	F5	Present to audience

Impress: Similar — Normal, Slide Sorter, Slide Show (bottom tabs or View menu).

4. Creating and Managing Slides

New slide

- Home → New Slide (Ctrl+M)
- Choose **layout**: Title Slide, Title and Content, Two Content, Blank, Comparison

Slide operations

- **Duplicate**: Right-click thumbnail → Duplicate
- **Delete**: Select slide → Delete key
- **Reorder**: Drag in left pane or Slide Sorter view

Text and lists

- **Title placeholder** — main heading (one per slide)
- **Content placeholder** — bullets and sub-bullets
- **Tab / Shift+Tab** — increase/decrease list level

6×6 rule: About 6 bullets max, ~6 words each — keeps slides readable.

5. Design and Themes

PowerPoint

- **Design tab** → **Themes** — colours, fonts, effects
- **Variants** — colour variations
- **Design Ideas** (Microsoft 365) — AI layout suggestions
- **Slide Size:** Standard (4:3) or Widescreen (16:9)

Impress

- **Slide** → **Apply Layout** / **Master Slide** for consistent branding
- **Format** → **Slide Design** or sidebar for themes

Background

- Design → Format Background — solid colour, gradient, or picture
 - Keep contrast high — dark text on light background (or reverse)
-

6. Insert Objects

Object	Use on slide
Pictures	Photos, logos — resize, crop
Shapes	Arrows, boxes for diagrams
Icons / SmartArt	Process flows, org charts (PowerPoint SmartArt)
Chart	Data from Excel/Calc
Table	Small tabular data
Video / Audio	Demos, background music (use sparingly)

Impress: Insert menu mirrors most options; charts link to Calc data.

7. Transitions (Between Slides)

PowerPoint: Transitions tab | **Impress:** Slide Transition deck

- Apply to one slide or **Apply To All**
- Set **duration** (speed)
- **Advance:** On mouse click or after X seconds

CCC tip: Use simple effects (Fade, Push, Wipe) — avoid distracting spins in formal talks.

8. Animations (On-Slide Objects)

PowerPoint: Animations tab | **Impress:** Custom Animation

Type	Example
Entrance	Appear, Fly In, Fade
Emphasis	Pulse, Grow/Shrink
Exit	Disappear
Motion path	Move along path

Start options: On Click / With Previous / After Previous

Animation Pane: Control order of multiple effects

Example: Bullet list — appear one by one on click.

9. Slide Show Delivery

Action	Key / command
Start from beginning	F5
Start from current slide	Shift+F5
Next slide	Click, Space, Enter, →
Previous slide	← or Backspace
Black screen	B
End show	Esc
Pen (draw on slide)	Ctrl+P (PowerPoint)

Presenter View (PowerPoint): Slide Show → Use Presenter View — notes, timer, next slide preview on your screen only.

10. Print and Export

- **Print:** Full slides, **handouts** (2 or 6 per page), notes pages
 - **PDF:** File → Save As / Export as PDF — sharing and printing
 - **Video:** File → Export → Create Video (.mp4) — PowerPoint
 - **Slideshow file:** `.ppsx` — opens directly in slide show mode (PowerPoint)
-

11. Sample Presentation Outline (CCC Topic)

Title: Digital India / Computer Literacy / Cyber Awareness

Slide #	Content
---------	---------

1	Title slide — topic, your name, institute
2	Introduction — what is the topic?
3–4	Key points (bullets, one idea per slide)
5	Chart or SmartArt — statistics or process
6	Image slide — relevant photo with caption
7	Summary — 3–4 takeaway points
8	Thank you — contact / questions

Lab Tasks

1. Create **8-slide** presentation: "Digital India" (or teacher-assigned topic)
 2. Apply one theme consistently in PowerPoint; repeat in Impress with equivalent theme
 3. Use Title + Content layouts; no overcrowded slides (6×6 rule)
 4. Add **transitions** to all slides (same simple effect)
 5. Animate bullet list — appear one by one on click
 6. Insert picture and one chart or SmartArt/diagram
 7. Run slideshow with F5; practice B (black screen) and Esc
 8. Export as **PDF** from both PowerPoint and Impress
 9. Optional: Export short video or save **.ppsx** slideshow file
-

Summary Checklist

- ☐ I can create slides in PowerPoint and Impress
 - ☐ I apply themes and consistent layouts
 - ☐ I know transition vs animation difference
 - ☐ I insert images, charts, and shapes
 - ☐ I present with F5 and keyboard navigation
 - ☐ I export PDF (and optional video/ppsx)
-

Quick MCQs (10 Questions)

1. LibreOffice presentation software is:

- a) Writer
- b) Calc
- c) Impress
- d) Draw only

2. Shortcut to start slideshow from beginning:

- a) F1
- b) F5
- c) Ctrl+S

- d) Alt+F5

3. Transition applies to:

- a) Text colour inside a shape
- b) Movement between slides
- c) Spell check
- d) Printer setup

4. Animation applies to:

- a) Objects on a slide
- b) Only the first slide
- c) File name
- d) Operating system

5. Default PowerPoint file extension:

- a) .docx
- b) .pptx
- c) .xlsx
- d) .odp

6. Impress default file extension:

- a) .pptx
- b) .odp
- c) .ods
- d) .pdf

7. 6×6 guideline suggests:

- a) 6 slides only per presentation
- b) Roughly 6 bullets with short text per slide
- c) 6 fonts per slide
- d) 6 hours per slide

8. Shift+F5 starts slideshow:

- a) From first slide always
- b) From current slide
- c) In black screen only
- d) Without keyboard

9. Handout printing means:

- a) Multiple small slides per printed page
- b) Printing only animations
- c) Printing BIOS
- d) One slide per workbook

10. .ppsx file type:

- a) Word document
- b) Opens PowerPoint directly in slideshow mode

- c) Excel macro
- d) Linux kernel

MCQ Answers

1-c | 2-b | 3-b | 4-a | 5-b | 6-b | 7-b | 8-b | 9-a | 10-b

Subject 6: Internet & Web Browsing (8 Hours)

Learning Objectives

- Explain internet, WWW, browser, URL, and ISP
 - Connect to the internet using Wi-Fi and mobile data
 - Browse websites safely using HTTPS and bookmarks
 - Use search engines effectively for CCC exam topics
 - Download and upload files correctly
 - Understand cookies, cache, and browser privacy settings
 - Follow safe browsing habits to avoid scams and malware
-

1. Internet Basics

Internet = a global network (vishwa jaal) of millions of computers connected through cables, fibre, and wireless links.

WWW (World Wide Web) = a system of linked web pages on the internet, accessed through browsers using HTTP/HTTPS.

Term	Meaning
Website	Collection of related web pages on one server
Web page	Single HTML document shown in browser
Home page	First/main page of a website
Hyperlink	Clickable link to another page or site
Server	Computer that stores and sends web pages
Client	Your computer or phone that requests pages

How you connect to the internet

1. **ISP (Internet Service Provider)** — company that gives internet (Jio, Airtel, BSNL, local broadband)
2. **Modem** — converts signal from ISP line to digital data
3. **Router** — shares connection to many devices (often Wi-Fi)
4. **Wi-Fi** — wireless connection within home/office
5. **Mobile data (4G/5G)** — internet through SIM on phone

IP Address — unique number of a device on the internet (like house number).

Domain name — easy name like `google.com` instead of remembering IP numbers.

DNS (Domain Name System) — converts domain name to IP address.

2. Web Browser

A **browser** is software to open and view websites.

Popular browsers (CCC exam)

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari (Apple devices)

Parts of browser window

Part	Use
Address bar	Type URL or search keywords
Tabs	Open many pages (Ctrl+T = new tab)
Back / Forward	Visit previous/next page
Refresh (F5)	Reload current page
Bookmarks / Favorites	Save site for quick access (Ctrl+D)
History (Ctrl+H)	List of pages you visited
Downloads (Ctrl+J)	Files saved from internet
Settings	Privacy, homepage, default search engine

Browser settings students should know

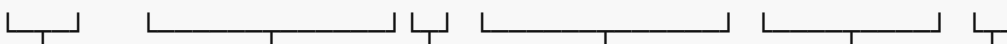
- Set **homepage** (page that opens on start)
 - Clear **cache** and **cookies** when site behaves wrongly
 - Enable **pop-up blocker** for safer browsing
 - Turn on **safe browsing** warnings (Chrome/Edge)
-

3. URL — Uniform Resource Locator

URL = full web address of a page or file.

Structure of a URL

`https://www.example.gov.in:443/courses/index.html?page=1#top`



protocol	domain	port	path	query	fragment
----------	--------	------	------	-------	----------

Part	Example	Meaning
Protocol	https://	Rules for connection
Domain	www.example.gov.in	Website name
Path	/courses/index.html	Location of page on server
Query	?page=1	Extra parameters

HTTP vs HTTPS

Protocol	Security	Use
HTTP	Not encrypted	General reading only
HTTPS	Encrypted (SSL/TLS)	Login, banking, payments

HTTPS shows a **padlock icon** in address bar. Always use HTTPS for passwords and online payment.

Common top-level domains (TLD)

- **.com** — commercial
- **.org** — organizations
- **.edu** — educational
- **.gov.in** — Indian government (official)
- **.nic.in** — National Informatics Centre (government)

4. ISP and Connection Types

ISP provides internet access for a fee (monthly plan / prepaid recharge).

Connection type	Speed	Typical use
Broadband (DSL/Fibre)	High	Home, office
Mobile data	Medium-high	Phone, hotspot
Dial-up (old)	Very slow	Rare today

Data usage terms

- **Bandwidth** — amount of data that can flow per second (Mbps)
 - **Data limit** — GB allowed per month in mobile plan
 - **Hotspot** — share phone internet to laptop via Wi-Fi
-

5. Search Engines

Search engine finds web pages matching your keywords.

Popular search engines

Google, Bing, DuckDuckGo, Yahoo

Effective search tips (CCC useful)

- Use **specific keywords**: NIELIT CCC syllabus 2025 not just computer
- **Quotes** for exact phrase: "Digital India mission"
- **site:** filter: site:gov.in UMANG app
- **filetype:** filter: filetype:pdf CCC study material
- Open results from **trusted sources** (.gov.in, .edu, official institute sites)

Evaluating search results

- Check **date** of article — old news may be wrong
 - Avoid clickbait titles
 - Cross-check important facts on official website
-

6. Download and Upload

Action	Direction	Example
Download	Internet → your device	Save PDF, image, app
Upload	Your device → internet	Attach file to email, cloud upload

Safe download steps

1. Download only from **official or trusted** websites
2. Check file **extension** (.pdf, .docx, .jpg) — avoid unknown .exe unless sure
3. Scan with **antivirus** after download
4. Note **save location** (Downloads folder)
5. Do not download **pirated software** or cracked games (virus risk)

Upload uses

- Email attachments
 - Google Drive / OneDrive
 - Online form document upload (admission, job application)
-

7. Cookies, Cache, and Sessions

Cookies

Small text files stored by browser on your computer. Websites use cookies to:

- Remember login (session cookie)

- Save preferences (language, theme)
- Track usage (advertising cookies)

Risk: Third-party cookies can track browsing. Clear cookies or use private/incognito mode on shared PC.

Cache

Temporary copies of images and pages stored locally for **faster loading**.

If a website shows old content, clear cache and refresh.

Incognito / Private browsing

- Does not save history on local PC (still visible to ISP/school network admin)
- Useful on **cyber café** or library computer
- Always **log out** after banking/email on shared PC

8. Internet Safety While Browsing

Common online risks

Risk	What happens	Protection
Phishing site	Fake login page steals password	Check URL, padlock, official domain
Malware download	Virus in fake software	Official app stores only
Fake offers	"Free iPhone" scam links	Do not click suspicious ads
Public Wi-Fi sniffing	Data intercepted	Avoid banking on open Wi-Fi

Safe browsing checklist

- Use **HTTPS** for sensitive sites
- Keep **browser and Windows updated**
- Install **antivirus** and enable firewall
- Do not share **OTP, PIN, password** on call or chat
- Verify **.gov.in** for government schemes
- Report fraud at **cybercrime.gov.in**

Parental and student discipline

- Limit screen time (samay prabandhan)
- Do not meet strangers from chat without guardian
- Think before posting personal photos (digital footprint)

9. Important CCC Exam Terms

Term	Full form / meaning
WWW	World Wide Web
URL	Uniform Resource Locator

ISP	Internet Service Provider
HTTP	HyperText Transfer Protocol
HTTPS	HTTP Secure
DNS	Domain Name System
IP	Internet Protocol address
Wi-Fi	Wireless Fidelity
HTML	HyperText Markup Language (web page language)

Lab Tasks

1. **Connect:** Connect laptop/phone to Wi-Fi or mobile hotspot; note ISP name and signal strength.
2. **Browser tour:** Open Chrome or Edge — identify address bar, tabs, bookmarks, history, downloads.
3. **URL practice:** Visit `https://www.digitalindia.gov.in` — note protocol, domain, padlock icon.
4. **Search:** Search `NIELIT CCC exam pattern` — open one official/trusted result; bookmark it (Ctrl+D).
5. **Advanced search:** Use `site:gov.in DigiLocker` and open official page.
6. **Download:** Download a PDF from a government site; save to `Documents\CCC` folder.
7. **Upload:** Upload the same PDF to Google Drive (or institute cloud); note share link steps.
8. **Privacy:** Open browser settings — find option to clear cookies and cache; list 3 cookie uses.
9. **HTTPS check:** Compare HTTP vs HTTPS site — observe padlock difference.
10. **Safety list:** Write 5 signs of a fake/phishing website.

Summary Checklist

- ☐ I can explain internet, WWW, browser, URL, and ISP
- ☐ I know parts of browser and shortcuts (Ctrl+T, Ctrl+D, Ctrl+H, Ctrl+J)
- ☐ I can read URL parts and difference between HTTP and HTTPS
- ☐ I can search effectively using keywords and site: filter
- ☐ I can download and upload files safely
- ☐ I understand cookies, cache, and private browsing
- ☐ I follow safe browsing rules for CCC and daily life

Quick-Check MCQs (Questions Only)

1. Internet is best described as (a) a global network of connected computers (b) a single supercomputer in Delhi (c) only email system (d) a type of printer
2. WWW stands for (a) World Wide Web (b) World Windows Work (c) Wide Web Wire (d) Web World Write
3. Software used to open websites is called (a) browser (b) spreadsheet (c) compiler (d) scanner driver
4. URL is (a) the address of a web page (b) a type of RAM (c) printer command (d) keyboard layout
5. ISP provides (a) internet connection service (b) free MS Office license (c) CPU replacement (d) monitor repair

6. HTTPS compared to HTTP is (a) more secure with encryption (b) slower and never used (c) only for games (d) without any difference
 7. Download means (a) saving a file from internet to your device (b) sending file from device to internet (c) deleting browser history (d) shutting down the PC
 8. Cookies in a browser are (a) small files websites store on your computer (b) physical biscuits for lab (c) antivirus programs (d) types of CPU
 9. To search only government India sites, useful operator is (a) site:gov.in (b) print:gov (c) delete:all (d) virus:scan
 10. Padlock icon in address bar usually indicates (a) secure HTTPS connection (b) website is always free (c) no images on page (d) computer has virus
-

Word Processing — Practical Lab Pack (CCC)

Software: MS Word 2016+ or LibreOffice Writer

Total labs: 6 · **Time:** ~12 hours practical

Lab 1: Basic Document (2 hrs)

1. Open blank document
2. Title "**My Introduction**" — Heading 1, centre align
3. Three paragraphs about yourself — justify, 1.5 line spacing
4. Save as `Lab1_Introduction.docx` (or `.odt`)

Check: Font size 12 body, title larger, file opens correctly

Lab 2: Formatting (2 hrs)

1. Open Lab 1 file
 2. Bold your name; italic your school/institute; underline email
 3. Apply bullet list of 5 hobbies
 4. Numbered list of 3 goals for CCC course
 5. Save As PDF
-

Lab 3: Table (2 hrs)

1. Create **4×4 table** — Student Details
 2. Rows: Name, Address, Phone, Email (Field | Detail)
 3. Shade header row; borders all cells
 4. Save as `Lab3_StudentTable.docx`
-

Lab 4: Formal Letter (2 hrs)

1. Create one-page **application letter** to Principal/Director
2. Subject: Admission to DCA/CCC course

3. Date top right; proper greeting; signature line
 4. Header: institute name; footer: page number
 5. Save + print preview
-

Lab 5: Resume (2 hrs)

1. Use template or create from scratch
 2. Sections: Objective, Education, Skills, Contact
 3. Insert small photo (optional) with wrap text
 4. Export PDF for submission practice
-

Lab 6: Mail Merge (2 hrs)

1. Create Excel list: 5 names + cities
 2. Word/Writer mail merge letter: "Dear «Name», Greetings from «City»..."
 3. Preview and generate 5 letters
 4. Print or save merged PDF
-

Trainer Sign-off Sheet

Lab	Student Name	Date	Initial
1			
2			
3			
4			
5			
6			

Spreadsheet — Practical Lab Pack (CCC)

Software: MS Excel or LibreOffice Calc

Total labs: 6 · **Time:** ~12 hours practical

Lab 1: Data Entry (2 hrs)

1. Create workbook `CCC_Marks.xlsx`
 2. Columns: Name, Roll, Sub1, Sub2, Sub3
 3. Enter 8 students with sample marks (0–100)
 4. Format header row bold, centre
-

Lab 2: Formulas (2 hrs)

1. Column **Total** = sum of 3 subjects
 2. Column **Average** = average of 3 subjects
 3. Column **Result** = IF average ≥ 40 , "Pass", else "Fail"
 4. Verify by changing one mark — formulas update
-

Lab 3: Functions (2 hrs)

1. Below table calculate:
 - Highest marks (MAX)
 - Lowest marks (MIN)
 - Class average (AVERAGE)
 - Count pass students (COUNTIF)
 2. Label each result cell
-

Lab 4: Sort & Filter (1.5 hrs)

1. Sort table by Total descending
 2. Enable filter — show only Pass students
 3. Filter one subject marks > 75
-

Lab 5: Conditional Formatting (1.5 hrs)

1. Highlight marks < 40 in red
 2. Highlight marks ≥ 75 in green
 3. Data bars on Total column (optional)
-

Lab 6: Chart (3 hrs)

1. Column chart — student names vs Total marks
 2. Pie chart — Pass vs Fail count
 3. Add chart titles and axis labels
 4. Set print area; repeat header row; print preview
-

Trainer Sign-off Sheet

Lab	Student Name	Date	Initial
1–6			

Presentation — Practical Lab Pack (CCC)

Software: MS PowerPoint or LibreOffice Impress

Total labs: 4 · **Time:** ~8 hours practical

Lab 1: 5-Slide "About Yourself" (3 hrs)

Create presentation with **exactly 5 slides**:

Slide	Content
1	Title: Your Name — CCC Student (Title Slide layout)
2	About Yourself — photo + 3 bullet points
3	Education — school/college, course
4	Skills — computer skills list
5	Career Goal + Thank You

Apply one **theme** to all slides.

Lab 2: Transitions & Animation (2 hrs)

1. Same presentation — add **same transition** to all slides
 2. Animate bullet points on slide 2 — appear one by one
 3. Run slideshow F5; practice Esc to exit
-

Lab 3: SmartArt / Image (1.5 hrs)

1. New slide with **SmartArt** process diagram (3 steps: Learn → Practice → Pass CCC)
 2. Insert image from device with caption
 3. Save as PDF
-

Lab 4: Handouts & Export (1.5 hrs)

1. Print preview — **6 slides per page** handout
 2. Save as **.ppsx** (or Impress slideshow export)
 3. Export video/mp4 if software supports
-

Trainer Sign-off Sheet

Lab	Student Name	Date	Initial
1–4			

Practice — Module 01: Introduction to Computer

Section A — MCQ (50 questions)

1. Full form of CPU? a) Central Processing Unit b) Computer Personal Unit c) Control Processing Unit d) Central Process Unit
2. Brain of the computer? a) CPU b) Monitor c) Keyboard d) Printer
3. Which memory is temporary? a) RAM b) ROM c) Hard disk d) DVD
4. Which memory stores boot instructions? a) ROM b) RAM c) Cache d) Register
5. Input device? a) Keyboard b) Monitor c) Printer d) Speaker
6. Output device? a) Printer b) Mouse c) Scanner d) Microphone
7. 1 KB equals? a) 1024 bytes b) 1000 bytes c) 1024 bits d) 8 bytes
8. Binary uses digits? a) 0 and 1 b) 0 to 9 c) 1 to 9 d) A to F
9. Software example? a) MS Word b) CPU c) Monitor d) Keyboard
10. Hardware example? a) Keyboard b) Windows c) Chrome d) Antivirus
11. Pen drive is? a) Storage device b) Input device c) Output device d) ALU
12. Full form of RAM? a) Random Access Memory b) Read Access Memory c) Run All Memory d) Random Active Memory
13. Full form of ROM? a) Read Only Memory b) Run Only Memory c) Random Only Memory d) Read Open Memory
14. Supercomputer used for? a) Weather research b) Typing letters c) Email only d) Printing bills
15. Data processed into meaningful form is? a) Information b) Hardware c) Input d) Output
16. Which is system software? a) Operating System b) MS Excel c) Chrome d) Tally
17. ALU is part of? a) CPU b) Monitor c) Printer d) Mouse
18. VDU stands for? a) Visual Display Unit b) Very Digital User c) Video Disk Unit d) Virtual Data Unit
19. Multitasking means? a) Running many programs together b) One program only c) No OS d) Shutdown
20. Virus is? a) Harmful software b) Hardware part c) Input device d) Storage
21. Smartphone is a? a) Mobile computer / IT gadget b) Only phone c) Printer d) Scanner
22. Touchscreen is both? a) Input and output b) Only input c) Only storage d) Only CPU
23. Evolution: first generation used? a) Vacuum tubes b) Transistors c) IC chips d) AI processors
24. Fourth generation used? a) Microprocessors / VLSI b) Vacuum tubes c) Punch cards only d) Abacus
25. SSD compared to HDD is generally? a) Faster b) Slower c) Same speed d) Not storage
26. Bit is? a) Smallest unit of data b) 8 bytes c) 1 KB d) 1 MB
27. 8 bits = ? a) 1 Byte b) 1 KB c) 1 bit d) 1 GB
28. Motherboard function? a) Connects all components b) Print documents c) Browse internet d) Send email

29. Scanner is? a) Input device b) Output device c) CPU d) OS
30. Webcam is? a) Input device b) Output only c) Storage d) Printer
31. Bluetooth in laptop is for? a) Wireless connectivity b) Printing only c) Storage only d) Cooling
32. Tablet is an IT gadget for? a) Portable computing b) Only calls c) Only printing d) Only scanning
33. Application software does? a) User-specific tasks b) Boot computer only c) Connect wires d) Cool CPU
34. Firmware is? a) Software in hardware b) Pure hardware c) Virus d) Browser
35. Cold boot starts from? a) Power off state b) Sleep only c) Lock screen d) Hibernate only
36. GUI means? a) Graphical User Interface b) General User Internet c) Graphic Utility Input d) Global User ID
37. File extension .docx is for? a) Word document b) Excel c) PowerPoint d) Video
38. Backup purpose? a) Protect data from loss b) Delete files c) Slow PC d) Print faster
39. Microcomputer example? a) Laptop b) Mainframe c) Supercomputer only d) None
40. Cache memory is? a) Very fast small buffer b) Slowest storage c) Printer buffer only d) Email storage
41. Which enters data into computer? a) Input device b) Output device c) Monitor d) Speaker
42. CPU speed measured in? a) GHz b) Kg c) Inch d) Litres
43. Decimal 5 in binary? a) 101 b) 110 c) 111 d) 100
44. Binary 1010 in decimal? a) 10 b) 8 c) 12 d) 5
45. Mainframe used in? a) Banks and railways b) Home gaming only c) Calculator only d) Watch only
46. Headphones are? a) Output device b) Input device c) Storage d) CPU
47. Joystick is used for? a) Games / input b) Printing c) Storage d) Booting
48. Digital camera captures? a) Images as digital data b) Only sound c) Only text d) Only programs
49. Mobile app runs on? a) Smartphone/tablet b) Printer only c) Scanner only d) Mouse only
50. E-governance uses computers for? a) Online citizen services b) Only games c) Only music d) Only photos

Section B — Short Answer (15)

1. Define key concept in Introduction to Computer.
2. List five important terms from Introduction to Computer.
3. Name three practical tasks for Introduction to Computer.
4. Why is Introduction to Computer important for CCC exam?
5. Give two exam tips for Introduction to Computer.
6. Define key concept in Introduction to Computer.
7. List five important terms from Introduction to Computer.
8. Name three practical tasks for Introduction to Computer.
9. Why is Introduction to Computer important for CCC exam?
10. Give two exam tips for Introduction to Computer.

11. Define key concept in Introduction to Computer.
12. List five important terms from Introduction to Computer.
13. Name three practical tasks for Introduction to Computer.
14. Why is Introduction to Computer important for CCC exam?
15. Give two exam tips for Introduction to Computer.

Section C — Practical (8)

1. Complete lab task 1 from Introduction to Computer study notes.
 2. Complete lab task 2 from Introduction to Computer study notes.
 3. Demonstrate skill to trainer on PC.
 4. Screenshot or checklist signed by trainer.
 5. Revise wrong MCQs from practice test.
 6. Peer review exercise in lab.
 7. Timed 10-minute drill on key shortcuts.
 8. Document steps in notebook.
-

Practice — Module 02: Operating System

Section A — MCQ (50 questions)

1. Windows is a? a) Operating System b) Browser c) Spreadsheet d) Antivirus
2. Linux is? a) Open source OS b) Closed browser c) Word processor d) Printer driver
3. OS function includes? a) File management b) Only gaming c) Only printing d) Only email
4. Shortcut Copy? a) Ctrl+C b) Ctrl+V c) Ctrl+X d) Ctrl+Z
5. Shortcut Paste? a) Ctrl+V b) Ctrl+C c) Ctrl+S d) Ctrl+P
6. Shortcut Save? a) Ctrl+S b) Ctrl+P c) Ctrl+N d) Ctrl+O
7. Win+E opens? a) File Explorer b) Excel c) Edge only d) Email
8. Win+L? a) Lock computer b) Log off always c) Open Word d) Print
9. F2 key? a) Rename b) Refresh c) Save d) Print
10. Delete sends file to? a) Recycle Bin b) Permanent delete always c) Desktop d) Network
11. Shift+Delete? a) Permanent delete b) Recycle Bin c) Copy d) Rename
12. Task Manager opens with? a) Ctrl+Shift+Esc b) F5 c) Alt+H d) Win+P
13. Default drive letter for Windows? a) C: b) A: c) Z: d) M:
14. Folder contains? a) Files and subfolders b) Only CPU c) Only RAM d) Nothing
15. Ctrl+X is? a) Cut b) Copy c) Paste d) Undo
16. Ctrl+Z is? a) Undo b) Redo c) Save d) Print

17. Alt+Tab? a) Switch windows b) Close all c) Shutdown d) Save
18. Zip folder purpose? a) Compress files b) Delete files c) Print files d) Email only
19. Desktop wallpaper changed via? a) Personalize/Settings b) BIOS only c) Notepad d) Calculator
20. Uninstall program via? a) Settings → Apps b) Delete .exe only c) Recycle Bin d) Paint
21. GUI OS uses? a) Icons and mouse b) Only commands c) No monitor d) Telegraph
22. File path shows? a) Location of file b) CPU speed c) RAM size d) Printer ink
23. Invalid filename character? a) * b) _ c) - d) 0
24. Snipping Tool for? a) Screenshot b) Virus scan c) Spreadsheet d) Database
25. Administrator account has? a) Full system control b) No install rights c) Guest only d) No login
26. Linux mascot/tradition often? a) Penguin Tux b) Apple only c) Windows logo d) None
27. Install program on Windows? a) Run installer / Store / setup.exe b) Delete folder only c) Rename file d) F5
28. Remove program properly? a) Uninstall from Settings b) Delete shortcut only c) Hide icon d) Lock PC
29. System tray shows? a) Clock, network, volume b) CPU chip c) Keyboard keys d) Hard disk
30. Start menu used to? a) Launch programs b) Format disk only c) Destroy data d) Remove OS
31. Ctrl+A? a) Select all b) Save all c) Print all d) Delete all
32. Copy vs Move: Copy? a) Original remains b) Original deleted c) Both deleted d) Neither exists
33. Restore deleted file from? a) Recycle Bin b) CPU c) Monitor d) Keyboard
34. Hibernate saves session to? a) Disk then low power b) RAM only always c) Cloud only d) Printer
35. Windows Update purpose? a) Security patches b) Delete files c) Remove OS d) Slow internet
36. Pen drive appears as drive letter when? a) Plugged in USB b) Always C: c) Never d) Only on Linux
37. Search in File Explorer finds? a) Files in folder/search scope b) Only internet c) Only emails d) BIOS
38. Notepad creates? a) Plain text files b) Excel sheets c) Database d) Video
39. Control Panel / Settings for? a) System configuration b) Games only c) Hardware repair physical d) Cable only
40. Shortcut to desktop? a) Send to Desktop shortcut b) Delete file c) Format C: d) Uninstall OS
41. Safe eject USB why? a) Prevent data corruption b) Speed up CPU c) Add RAM d) Install OS
42. Linux distributions include? a) Ubuntu, Fedora b) Word, Excel only c) Chrome only d) None
43. Open source means? a) Source code can be studied/modified b) Always free hardware c) No internet d) No GUI
44. Taskbar location usually? a) Bottom of screen b) Inside CPU c) On printer d) In BIOS
45. File extension identifies? a) File type b) CPU model c) RAM size d) User name

46. Win+D shows? a) Desktop b) Delete files c) Download d) Database
47. Increase list indent in Explorer sort? a) Sort by name/date/size b) Print c) Scan d) Email
48. Lock PC when away using? a) Win+L b) Delete c) F5 d) Ctrl+P
49. Guest account typically? a) Limited permissions b) Full admin c) No login ever d) Root on Linux always
50. Antivirus on Windows often? a) Windows Defender built-in b) Not available c) Part of CPU d) Part of mouse

Section B — Short Answer (15)

1. Define key concept in Operating System.
2. List five important terms from Operating System.
3. Name three practical tasks for Operating System.
4. Why is Operating System important for CCC exam?
5. Give two exam tips for Operating System.
6. Define key concept in Operating System.
7. List five important terms from Operating System.
8. Name three practical tasks for Operating System.
9. Why is Operating System important for CCC exam?
10. Give two exam tips for Operating System.
11. Define key concept in Operating System.
12. List five important terms from Operating System.
13. Name three practical tasks for Operating System.
14. Why is Operating System important for CCC exam?
15. Give two exam tips for Operating System.

Section C — Practical (8)

1. Complete lab task 1 from Operating System study notes.
 2. Complete lab task 2 from Operating System study notes.
 3. Demonstrate skill to trainer on PC.
 4. Screenshot or checklist signed by trainer.
 5. Revise wrong MCQs from practice test.
 6. Peer review exercise in lab.
 7. Timed 10-minute drill on key shortcuts.
 8. Document steps in notebook.
-

Practice — Module 03: Word Processing

Section A — MCQ (50 questions)

1. Word file extension? a) .docx b) .xlsx c) .pttx d) .accdb
2. LibreOffice Writer extension? a) .odt b) .docx only c) .xls d) .ppt
3. Bold shortcut? a) Ctrl+B b) Ctrl+I c) Ctrl+U d) Ctrl+P
4. Italic shortcut? a) Ctrl+I b) Ctrl+B c) Ctrl+S d) Ctrl+N

5. Underline shortcut? a) Ctrl+U b) Ctrl+B c) Ctrl+K d) Ctrl+L
6. Save shortcut? a) Ctrl+S b) Ctrl+P c) Ctrl+O d) F5
7. Page break shortcut? a) Ctrl+Enter b) Enter c) Shift+Enter d) Alt+Enter
8. Mail merge used for? a) Bulk personalized letters b) Virus scan c) Database query d) Print driver
9. Mail merge tab in Word? a) Mailings b) Insert only c) View only d) Draw only
10. Justify alignment? a) Both edges straight b) Center only c) Left only d) Right only
11. Find shortcut? a) Ctrl+F b) Ctrl+H c) Ctrl+G d) Ctrl+N
12. Replace shortcut? a) Ctrl+H b) Ctrl+R c) Ctrl+J d) Ctrl+K
13. Spell check key? a) F7 b) F5 c) F2 d) F12
14. Insert table from? a) Insert tab b) Review only c) Mailings only d) Help
15. Header/footer from? a) Insert tab b) Home font c) View only d) Draw
16. Page numbers inserted via? a) Insert → Page Number b) Home Bold c) Review d) File Close
17. Export Word to PDF? a) Save As PDF b) Cannot export c) Print only paper d) Insert PDF
18. Portrait orientation? a) Vertical page b) Horizontal page c) Circular d) None
19. Landscape orientation? a) Horizontal page b) Vertical page c) Square only d) None
20. Wrap text for image? a) Picture Format → Wrap Text b) Home font color c) Review d) Mailings
21. Heading styles used for? a) Document structure b) Virus removal c) Internet d) CPU
22. Line spacing changed in? a) Home → Paragraph b) Insert Picture c) Review d) Draw
23. Tab in table moves to? a) Next cell b) New document c) Internet d) Shutdown
24. Merge cells? a) Combine cells into one b) Delete table c) Print d) Save PDF
25. Template file type often? a) .dotx b) .mp4 c) .exe d) .zip only
26. Ctrl+P in Word? a) Print b) Paste c) Paragraph d) Picture
27. Bullets in list from? a) Home tab b) Insert Chart c) References only d) Mailings only
28. Page margins set in? a) Layout tab b) Insert c) Review d) Draw
29. Columns in document? a) Layout → Columns b) Home font c) Insert table only d) None
30. Thesaurus shortcut? a) Shift+F7 b) F1 c) F12 d) Ctrl+T
31. Word count in? a) Review tab b) Insert c) Draw d) Help only
32. Clear formatting? a) Home → Clear All Formatting b) Delete key only c) BIOS d) Task Manager
33. Section break for? a) Different headers in sections b) Virus c) Email d) CPU upgrade
34. Ctrl+Z in Word? a) Undo b) Redo c) Zoom d) Cut

35. Hyperlink insert shortcut? a) Ctrl+K b) Ctrl+H c) Ctrl+L d) Ctrl+U
36. Footnote used for? a) Reference at page bottom b) Header text c) Footer page number only d) Mail merge
37. Resume created using? a) Word/Writer templates b) Excel only c) Access only d) Paint only
38. Mail merge data source often? a) Excel list b) Printer c) Scanner d) CPU
39. Print preview before? a) Printing to check layout b) Deleting file c) Formatting C: d) Uninstall
40. Double spacing means? a) 2.0 line spacing b) Two pages c) Two columns always d) Two files
41. Insert picture from? a) Insert → Pictures b) Review c) Mailings only d) References only
42. Writer is part of? a) LibreOffice b) Windows OS c) Chrome d) Antivirus
43. MS Word is? a) Word processor b) Spreadsheet c) Database d) Browser
44. Cut shortcut? a) Ctrl+X b) Ctrl+C c) Ctrl+V d) Ctrl+Z
45. Paste shortcut? a) Ctrl+V b) Ctrl+C c) Ctrl+X d) Ctrl+S
46. Select all? a) Ctrl+A b) Ctrl+S c) Ctrl+P d) Ctrl+F
47. Red underline in Word often means? a) Spelling error b) Saved file c) Printed d) Merged
48. Green underline may mean? a) Grammar suggestion b) Virus c) Network error d) Low ink
49. Envelopes printed via? a) Mailings / Envelopes b) Insert Chart c) Review d) Draw
50. Table of Contents needs? a) Heading styles b) Only bold text c) Images only d) Videos only

Section B — Short Answer (15)

1. Define key concept in Word Processing.
2. List five important terms from Word Processing.
3. Name three practical tasks for Word Processing.
4. Why is Word Processing important for CCC exam?
5. Give two exam tips for Word Processing.
6. Define key concept in Word Processing.
7. List five important terms from Word Processing.
8. Name three practical tasks for Word Processing.
9. Why is Word Processing important for CCC exam?
10. Give two exam tips for Word Processing.
11. Define key concept in Word Processing.
12. List five important terms from Word Processing.
13. Name three practical tasks for Word Processing.
14. Why is Word Processing important for CCC exam?
15. Give two exam tips for Word Processing.

Section C — Practical (8)

1. Complete lab task 1 from Word Processing study notes.
2. Complete lab task 2 from Word Processing study notes.

3. Demonstrate skill to trainer on PC.
 4. Screenshot or checklist signed by trainer.
 5. Revise wrong MCQs from practice test.
 6. Peer review exercise in lab.
 7. Timed 10-minute drill on key shortcuts.
 8. Document steps in notebook.
-

Practice — Module 04: Spreadsheet

Section A — MCQ (50 questions)

1. Excel extension? a) .xlsx b) .docx c) .pptx d) .odt
2. Calc is part of? a) LibreOffice b) Windows c) Chrome d) Word
3. Formula starts with? a) = b) + c) # d) @
4. SUM() used for? a) Addition/total b) Average c) Count text d) Print
5. AVERAGE() returns? a) Mean value b) Largest c) Smallest d) Text join
6. MAX() returns? a) Largest value b) Smallest c) Count d) Date
7. MIN() returns? a) Smallest value b) Largest c) Sum d) Average text
8. COUNT() counts? a) Numeric cells b) All text c) Blank only d) Sheets
9. Cell A1 is? a) Column A row 1 b) Row A column 1 c) Invalid d) Chart
10. Absolute reference? a) \$A\$1 b) A1 only always c) #REF d) SUM
11. IF function checks? a) Condition true/false b) Text color c) Print area d) Chart type
12. Pie chart best for? a) Parts of whole b) Time trend only c) Large database d) Email
13. Column chart for? a) Comparing categories b) Parts of whole only c) Email d) Boot
14. Line chart for? a) Trend over time b) Pie parts c) Database keys d) Virus scan
15. Sort data? a) Data tab → Sort b) Insert Chart c) Review d) Draw
16. Filter adds? a) Dropdown arrows in headers b) Virus c) RAM d) Printer
17. Conditional formatting? a) Format by rules b) Delete rows c) Install OS d) Send email
18. Print area sets? a) Range to print b) RAM size c) CPU d) Monitor
19. Print titles repeat? a) Header rows each page b) Footer only c) Delete data d) Chart only
20. Workbook contains? a) Worksheets b) Only one cell c) Only charts d) Emails
21. Worksheet is? a) Tab/sheet in workbook b) CPU c) Printer d) Browser
22. Fill handle used to? a) Copy formula/extend series b) Print c) Shutdown d) Email

23. Ctrl+1 opens? a) Format Cells b) Save c) Print d) New book
24. Merge cells? a) Combine into one cell b) Delete sheet c) Scan d) Boot
25. COUNTIF counts? a) Cells meeting criteria b) All sheets c) Printers d) Folders
26. SUMIF adds? a) Values matching criteria b) All text c) Charts d) BIOS
27. TODAY() returns? a) Current date b) Sum c) Average d) Text hello
28. ROUND(3.456,2) gives? a) 3.46 b) 3.45 c) 3.5 always wrong d) 3456
29. Percentage formula example? a) =B2/C2*100 b) =B2+C2 c) =SUM(text) d) =IF color
30. Remove Duplicates? a) Data tab b) Home font c) Insert d) View
31. 3D formula SUM(Sheet1:Sheet3!B2)? a) Sum B2 across sheets b) One cell typo c) Chart d) Email
32. Link to Sheet2 A1? a) =Sheet2!A1 b) =A1Sheet2 c) =LINK(A1) d) =WEB(A1)
33. Relative reference when copied down? a) Adjusts row/column b) Stays fixed always c) Deletes d) Prints
34. F4 in Excel toggles? a) Absolute/relative refs b) Save c) Print d) Close
35. Gridlines print option in? a) Page Layout b) Insert c) Review d) Mailings
36. Chart inserted from? a) Insert tab b) File Close c) BIOS d) Task Manager
37. Legend in chart shows? a) Series labels b) CPU c) RAM d) Password
38. #DIV/0! error means?
- a) Divide by zero b) Success c) Saved d) Printed
39. Currency format shows? a) ₹ or \$ amounts b) Only text c) Only dates d) Charts only
40. Bar chart similar to? a) Column chart horizontal b) Pie only c) Line only d) Email
41. AutoSum button does? a) Quick SUM formula b) Average only c) Print d) Save PDF
42. Freeze panes keeps visible? a) Header/label rows while scroll b) CPU c) Printer d) Desktop
43. Data validation restricts? a) Allowed cell input b) Internet c) OS install d) Email
44. Concatenate joins? a) Text strings b) Sheets c) Printers d) Folders
45. Ctrl+S in Excel? a) Save workbook b) Sum c) Sort d) Print
46. Active cell shown in? a) Name box / formula bar b) Taskbar only c) BIOS d) Printer
47. Shift+F11 adds? a) New worksheet b) New workbook always c) Chart d) Email
48. Calc compatible with? a) Excel formats mostly b) Only video c) Only audio d) Only PDF edit
49. Sorting Z to A is? a) Descending b) Ascending c) Random d) Print order
50. Excel extension? a) .xlsx b) .docx c) .pptx d) .odt

Section B — Short Answer (15)

1. Define key concept in Spreadsheet.
2. List five important terms from Spreadsheet.
3. Name three practical tasks for Spreadsheet.
4. Why is Spreadsheet important for CCC exam?
5. Give two exam tips for Spreadsheet.
6. Define key concept in Spreadsheet.
7. List five important terms from Spreadsheet.
8. Name three practical tasks for Spreadsheet.
9. Why is Spreadsheet important for CCC exam?
10. Give two exam tips for Spreadsheet.
11. Define key concept in Spreadsheet.
12. List five important terms from Spreadsheet.
13. Name three practical tasks for Spreadsheet.
14. Why is Spreadsheet important for CCC exam?
15. Give two exam tips for Spreadsheet.

Section C — Practical (8)

1. Complete lab task 1 from Spreadsheet study notes.
 2. Complete lab task 2 from Spreadsheet study notes.
 3. Demonstrate skill to trainer on PC.
 4. Screenshot or checklist signed by trainer.
 5. Revise wrong MCQs from practice test.
 6. Peer review exercise in lab.
 7. Timed 10-minute drill on key shortcuts.
 8. Document steps in notebook.
-

Practice — Module 05: Presentation

Section A — MCQ (50 questions)

1. PowerPoint extension? a) .pptx b) .docx c) .xlsx d) .odt
2. Impress is part of? a) LibreOffice b) Windows OS c) Chrome d) Excel
3. F5 starts slideshow from? a) Beginning b) Current slide only always c) End d) Print
4. Shift+F5 from? a) Current slide b) Beginning always c) Last slide d) Print preview
5. New slide shortcut? a) Ctrl+M b) Ctrl+N doc c) Ctrl+P d) F7
6. Transition applies to? a) Between slides b) Objects only c) Notes d) Handouts only
7. Animation applies to? a) Objects on slide b) Between slides c) Printer d) BIOS
8. Title slide layout has? a) Title + subtitle placeholders b) Blank only c) Excel table d) Database
9. Theme changes? a) Colors fonts effects b) CPU speed c) RAM d) Internet

10. Widescreen slide ratio often? a) 16:9 b) 4:3 only c) 1:1 only d) None
11. Duplicate slide? a) Right-click Duplicate b) F5 c) Delete d) Esc
12. Presenter View shows? a) Notes and timer b) BIOS c) Excel d) Email only
13. B key during show? a) Black screen b) Blue screen c) Boot d) Bold text
14. Esc ends? a) Slide show b) Windows c) Excel d) Internet
15. Handouts print? a) Multiple slides per page b) One slide only always c) Cannot print d) Email only
16. .ppsx opens in? a) Slideshow directly b) Word c) Access d) Notepad
17. Animation Pane controls? a) Order and timing b) Printer c) Virus d) RAM
18. SmartArt used for? a) Diagrams b) Spreadsheets c) Database d) Email server
19. Slide Sorter view for? a) Reorder slides b) Edit BIOS c) Format C: d) Scan
20. 6×6 rule means? a) Readable bullets ~6 words b) Excel rows c) RAM GB d) Print margin mm
21. Insert video from? a) Insert tab b) File Close c) BIOS d) Task Manager
22. Apply transition to all? a) Transitions → Apply To All b) Delete slides c) Save only d) Scan
23. Export presentation PDF? a) Save As PDF b) Cannot c) Print paper only d) Insert PDF
24. Slide Master edits? a) Master layout for all slides b) Notepad c) Calculator d) Paint only
25. Reading View? a) Full screen read slides b) Edit CPU c) Print only d) Email
26. Action button for? a) Hyperlink/navigation b) Virus scan c) RAM upgrade d) OS install
27. Best practice slides? a) Short bullets clear visuals b) Full essay paragraphs c) No title d) 50 animations each
28. Entrance animation? a) Object appears b) Object exits c) Slide changes d) Print starts
29. Design Ideas suggests? a) Layouts (365) b) CPU upgrades c) RAM d) Printer ink
30. Notes pane for? a) Speaker notes b) Student grades c) Database d) Email
31. Laser pointer in show? a) Ctrl+L in some versions / pen tool b) Delete slide c) Format disk d) Uninstall
32. Loop slideshow option? a) Repeats until Esc b) Never repeats c) Deletes file d) Emails slides
33. Hide slide? a) Skip in show but keep b) Delete permanently always c) Print only d) Email only
34. Rehearse timings records? a) Slide duration b) CPU temp c) RAM d) Ink level
35. Chart on slide data from? a) Excel often b) BIOS c) Registry only d) None
36. Compress media reduces? a) File size b) Slide count c) RAM d) Monitor size
37. Normal view is? a) Default edit view b) Print only c) Slideshow d) BIOS
38. Footer on slides can show? a) Date slide number b) CPU serial c) RAM only d) Password
39. Group objects? a) Right-click Group b) F5 c) Esc d) Delete OS

40. Impress file extension often? a) .odp b) .pptx only c) .doc d) .xls
41. Transition duration controls? a) Speed between slides b) CPU c) Internet d) Email size
42. Fly In is type of? a) Entrance animation b) Exit only c) Transition d) Print
43. Fade transition? a) Smooth appear between slides b) Delete slide c) Save PDF d) Mail merge
44. Thank you slide typically? a) Last slide b) First only c) Hidden always d) Printed only
45. About Yourself presentation has slides on? a) Intro education skills goal b) Only Excel c) Only Access d) Only BIOS
46. Ctrl+M in PowerPoint? a) New slide b) New document c) Mail merge d) Mock test
47. Slide layout Title and Content for? a) Bullets and title b) Blank only c) Database d) Spreadsheet edit
48. Export video mp4? a) File Export Create Video b) Insert c) Home d) Review
49. PowerPoint extension? a) .pptx b) .docx c) .xlsx d) .odt
50. Impress is part of? a) LibreOffice b) Windows OS c) Chrome d) Excel

Section B — Short Answer (15)

1. Define key concept in Presentation.
2. List five important terms from Presentation.
3. Name three practical tasks for Presentation.
4. Why is Presentation important for CCC exam?
5. Give two exam tips for Presentation.
6. Define key concept in Presentation.
7. List five important terms from Presentation.
8. Name three practical tasks for Presentation.
9. Why is Presentation important for CCC exam?
10. Give two exam tips for Presentation.
11. Define key concept in Presentation.
12. List five important terms from Presentation.
13. Name three practical tasks for Presentation.
14. Why is Presentation important for CCC exam?
15. Give two exam tips for Presentation.

Section C — Practical (8)

1. Complete lab task 1 from Presentation study notes.
 2. Complete lab task 2 from Presentation study notes.
 3. Demonstrate skill to trainer on PC.
 4. Screenshot or checklist signed by trainer.
 5. Revise wrong MCQs from practice test.
 6. Peer review exercise in lab.
 7. Timed 10-minute drill on key shortcuts.
 8. Document steps in notebook.
-

Practice — Module 06: Internet & Web Browsing

Section A — MCQ (50 questions)

1. Internet is? a) Global network of computers b) One PC c) Printer d) Keyboard
2. WWW stands for? a) World Wide Web b) World Web Wide c) Wide World Web d) Web World Wide
3. Browser example? a) Google Chrome b) MS Word c) Excel d) Access
4. URL stands for? a) Uniform Resource Locator b) Universal Reference Link c) Unified Resource Line d) User Resource Locator
5. ISP provides? a) Internet connection b) Word license c) Printer d) CPU
6. HTTPS indicates? a) Secure connection b) Slow site c) No images d) Free Wi-Fi
7. HTTP stands for? a) Hyper Text Transfer Protocol b) High Transfer Text Program c) Hyper Tool Transfer d) Home Text Protocol
8. Download means? a) Save from internet to PC b) Upload c) Delete d) Print
9. Upload means? a) Send from PC to internet b) Download c) Shutdown d) Format
10. Google is a? a) Search engine b) Operating system c) Word processor d) Printer
11. Bookmark saves? a) Favorite website b) Deletes site c) Formats PC d) Installs OS
12. Ctrl+T in browser? a) New tab b) Close tab c) Print d) Save page
13. History shows? a) Past visited pages b) RAM usage c) CPU model d) Printer queue
14. Home page is? a) First page of website b) Last page c) Error page d) BIOS
15. Web page is? a) Single HTML document b) Entire website always c) Email d) Database table
16. Modem/router connects? a) Home to ISP b) Keyboard to mouse c) Printer to scanner d) CPU to RAM only
17. Wi-Fi is? a) Wireless network b) Wired cable only c) Printer type d) Virus
18. Search with exact phrase use? a) Quotes "phrase" b) Brackets only c) Delete key d) F5 only
19. site:gov.in search limits to? a) Government sites b) All games c) All shops d) Printers
20. Pop-up blocker helps? a) Block unwanted windows b) Speed CPU c) Add RAM d) Print faster
21. Cookies store? a) Site preferences/login info b) Hardware drivers c) CPU firmware d) Printer ink
22. Refresh page? a) F5 or reload button b) Delete OS c) Format disk d) Uninstall browser
23. Address bar used to? a) Type URL or search b) Print c) Scan d) Save Excel
24. Browser tabs allow? a) Multiple pages open b) One page only c) No internet d) BIOS edit
25. Downloading from official source is? a) Safer practice b) Always virus c) Illegal always d) Impossible

26. Broadband is? a) High-speed internet b) Slow dial-up only c) Printer cable d) Keyboard type
27. Mobile data connects phone via? a) Cellular network b) Printer only c) Scanner only d) HDMI only
28. Hyperlink connects to? a) Another page/resource b) CPU only c) RAM only d) Printer ink
29. Domain .edu often? a) Educational institution b) Government India only c) Games only d) Printer
30. Phishing often via? a) Fake links/emails b) CPU upgrade c) RAM stick d) Mouse
31. Incognito/private mode? a) Less local history saved b) Fully anonymous always guaranteed c) Free internet d) No HTTPS
32. Search engine besides Google? a) Bing, DuckDuckGo b) Word, Excel c) Paint, Notepad d) None
33. Web server stores? a) Website files b) Printer ink c) Keyboard keys d) Mouse ball
34. HTML is? a) Markup language for web pages b) Spreadsheet c) Database d) OS
35. Downloading PDF from trusted site? a) Check HTTPS and domain b) Click any popup c) Share password d) Disable antivirus
36. Browser cache stores? a) Temporary page files b) CPU programs c) BIOS d) Printer drivers only
37. Clear cache can fix? a) Loading/display issues b) Broken keyboard c) No monitor d) No CPU
38. Default search engine set in? a) Browser settings b) BIOS only c) Word options d) Excel
39. FTP used for? a) File transfer b) Email only c) Word processing d) Presentation
40. Online form submit after? a) Reading instructions carefully b) Random clicks c) Sharing OTP d) Disabling HTTPS
41. Digital footprint means? a) Trail of online activity b) Physical shoe size c) Printer mark d) CPU serial
42. Web browser not same as? a) Search engine only b) Internet itself c) Website server d) All same always wrong
43. LAN connects? a) Local area computers b) Only printers globally c) Only phones satellite d) BIOS chips
44. DNS translates? a) Domain name to IP address b) Word to PDF c) Excel to PPT d) Email to print
45. IPv4 address looks like? a) 192.168.1.1 b) www.google.com c) user@gmail.com d) .docx
46. Streaming plays media? a) Without full download first b) Only after printing c) Only on paper d) Never online
47. Browser extension adds? a) Extra features b) CPU hardware c) RAM chips d) Printer toner
48. Safe download check? a) Verify publisher and HTTPS b) Double every popup c) Turn off firewall d) Share Aadhaar
49. Internet is? a) Global network of computers b) One PC c) Printer d) Keyboard
50. WWW stands for? a) World Wide Web b) World Web Wide c) Wide World Web d) Web World Wide

Section B — Short Answer (15)

1. Define key concept in Internet & Web Browsing.
2. List five important terms from Internet & Web Browsing.

3. Name three practical tasks for Internet & Web Browsing.
4. Why is Internet & Web Browsing important for CCC exam?
5. Give two exam tips for Internet & Web Browsing.
6. Define key concept in Internet & Web Browsing.
7. List five important terms from Internet & Web Browsing.
8. Name three practical tasks for Internet & Web Browsing.
9. Why is Internet & Web Browsing important for CCC exam?
10. Give two exam tips for Internet & Web Browsing.
11. Define key concept in Internet & Web Browsing.
12. List five important terms from Internet & Web Browsing.
13. Name three practical tasks for Internet & Web Browsing.
14. Why is Internet & Web Browsing important for CCC exam?
15. Give two exam tips for Internet & Web Browsing.

Section C — Practical (8)

1. Complete lab task 1 from Internet & Web Browsing study notes.
2. Complete lab task 2 from Internet & Web Browsing study notes.
3. Demonstrate skill to trainer on PC.
4. Screenshot or checklist signed by trainer.
5. Revise wrong MCQs from practice test.
6. Peer review exercise in lab.
7. Timed 10-minute drill on key shortcuts.
8. Document steps in notebook.